Roll Number: _____ Name_____ Group_____

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY, PATIALA
Department of Computer Science & Engineering
Computer Networks (UCS520) –Auxiliary Examination
Date: 08/09/2026 & Time: 5:00 PM **MM: 100 marks**

Faculty Name: Dr. Swati Sharma
Attempt/Answer all sub-parts like (a) and (b) for each question in one place. Do mention the Page number of your attempt on the front page of the answer sheet. Assume missing data (if any). Show all intermediate computations properly.

Q. No.		Question	Marks
Q1	(a)	A student is streaming a live online class, but the video frequently freezes. Using the OSI model, explain how different layers together contribute to this problem.	10
	(b)	Explain the concept of congestion control in computer networks. Differentiate between open-loop and closed-loop congestion control policies.	5
Q2	(a)	Describe the Go-Back-N ARQ protocol and highlight how it improves efficiency over Stop-and-Wait.	10
	(b)	Compare Multilevel TDM, Multiple-Slot TDM, and Pulse-Stuffed TDM with respect to their working principles.	5
Q3	(a)	Discuss the importance of subnetting. Assume an organization is granted the IP address block 192.168.0.0/16. The administrator wants to create 64 subnets. Solve the following: i. Find the subnet mask. ii. Find the number of addresses in each subnet. iii. Find the number of usable host addresses in each subnet. iv. Find the first and last addresses in subnet 1. v. Find the first and last addresses in subnet 64.	10
	(b)	Explain the process of address binding in IPv4 using ARP, and describe the steps involved in resolving an IP address to its corresponding MAC address.	5
Q4	(a)	A hospital network is designed to connect devices within a building, branches across a city, and servers with centralized storage. Short-range wireless devices are also supported. i. Identify and justify the appropriate network type (LAN, MAN, SAN, and PAN) for each requirement. ii. For communication between the emergency room and control center, identify the suitable data flow mode and explain the five basic components of data communication.	5+5
	(b)	Explain the purpose and significance of the following fields in the IPv4 header: i. Identification ii. Flags iii. Fragment Offset iv. Time to Live (TTL) v. Protocol	5
Q5	(a)	A signal with an initial power of 5 mW travels through a cable that has a loss of –0.25 dB/km. The cable length is 8 km. Calculate the final power of the signal at the receiver. Show all steps clearly using the decibel formula.	10
	(b)	Explain the difference between a hub and a switch in terms of collision domains, broadcast domains, and data transmission methods.	5
Q6	(a)	Which transport layer protocol is connection-oriented and which is connectionless? Explain briefly how each handles data transmission.	5+5
	(b)	Explain the role of Medium Access Control (MAC) protocols in a network. Compare Ethernet and Token Ring protocols, describe how they control access to the shared communication medium.	5
Q7		(i) Define SCTP and mention one advantage of SCTP over TCP and UDP. (ii) Explain the difference between Parity Bit and Cyclic Redundancy Check (CRC) as methods for error detection. Discuss their working principles, advantages, and limitations.	5+5